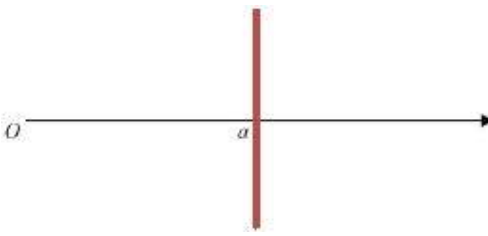
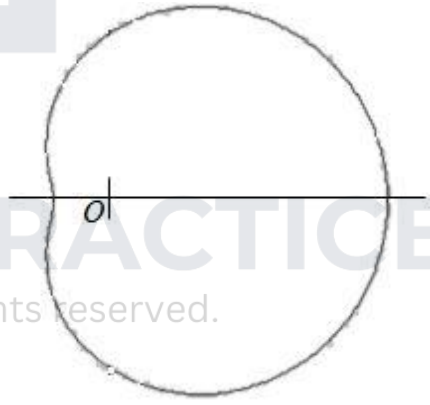


Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Draws correct line. Condone a half-line.	AO1.1a	M1	
Draws the full correct line and gives the correct intersection point on the initial line. Condone the angle omitted.	AO1.1b	A1	
Total 2 marks			

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	(i) States max value for r	AO1.1b	B1	Maximum value of $r = 5$
	(i) States min value for r	AO1.1b	B1	Minimum value of $r = 1$
(ii)	Draws simple closed curve enclosing pole	AO1.1a	M1	
	Draws correct shape with dimple (not cusp) when $\theta = \pi$	AO1.1b	A1	
(b)	Equates $3 + 2\cos\theta = 8\cos^2\theta$	AO1.1a	M1	$3 + 2\cos\theta = 8\cos^2\theta$ $8\cos^2\theta - 2\cos\theta - 3 = 0$
	Solves 'their' quadratic equation FT 'their' equation only if M1 has been awarded	AO1.1a	M1	$(4\cos\theta - 3)(2\cos\theta + 1) = 0$ $\cos\theta = \frac{3}{4}, \quad \cos\theta = -\frac{1}{2}$
	Obtains 2 values for θ for each value of $\cos\theta$ FT 'their' equation only if both M1 marks have been awarded	AO1.1b	A1F	$\theta = 0.723 \text{ or } \frac{2\pi}{3}$ $\theta = 5.56 \text{ or } \frac{4\pi}{3}$
	Substitutes 'their' $\cos\theta$ into	AO1.1a	M1	

a polar equation to find a value of r FT 'their' $\cos\theta$ only if both M1 marks have been awarded			$\cos\theta = \frac{3}{4} \Rightarrow r = \frac{9}{2}$ $\cos\theta = -\frac{1}{2} \Rightarrow r = 2$
Obtains both values of r correct for 'their' $\cos\theta$ values FT 'their' $\cos\theta$ only if both M1 marks have been awarded	AO1.1b	A1F	Intersection points $\left[\frac{9}{2}, 0.723\right]$, $\left[\frac{9}{2}, 5.56\right], \left[2, \frac{2\pi}{3}\right], \left[2, \frac{4\pi}{3}\right]$
Deduces that four values for θ exist and expresses points in required form	AO2.2a	R1	
Total 10 marks			

Q3.

(a) $4 \sin \theta (1 - \sin \theta) = 1$

Elimination of r or θ $\{r = 4[1 - (1/r)]\}$

M1

$$4 \sin^2 \theta - 4 \sin \theta + 1 = 0$$

$$\{r^2 - 4r + 4 = 0\}$$

A1

$$(2 \sin \theta - 1)^2 = 0 \Rightarrow \sin \theta = 0.5$$

*Valid method to solve quadratic eqn. $P1$
 $\{(r - 2)^2 = 0 \Rightarrow r = 2\}$*

m1

$$\theta = \frac{\pi}{6}, \theta = \frac{5\pi}{6}, r = 2$$

$$\left[P\left(2, \frac{\pi}{6}\right) \quad Q\left(2, \frac{5\pi}{6}\right)\right]$$

A1 for any two of the three.

A2,1

SC: Verification of $P\left(2, \frac{\pi}{6}\right)$ scores max

of B1 & a further B1 if $Q\left(2, \frac{5\pi}{6}\right)$ stated

(b) Area triangle $OPQ = \frac{1}{2} \times 2 \times r_Q \times \sin POQ$

*Any valid method to correct (ft eg on r_Q)
expression with just one remaining unknown*

M1

$$\text{Angle } POQ = \frac{5\pi}{6} - \frac{\pi}{6} \quad \left(= \frac{2\pi}{3} \right)$$

*Valid method to find remaining unknown
either relevant angle or relevant side*

m1

$$\text{Area triangle } OPQ = 2 \sin \frac{2\pi}{3} = \sqrt{3}$$

A1

Unshaded area bounded by line OP and arc $OP =$

$$\frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} [4(1 - \sin \theta)]^2 d\theta$$

*Use of $\frac{1}{2} \int r^2 d\theta$ for relevant area(s)
(condone missing/wrong limits)*

M1

$$= 8 \int (1 - 2 \sin \theta + \sin^2 \theta) d\theta$$

Correct expn of $(1 - \sin \theta)^2$

©2025 Exam Papers Practice. All rights reserved. B1

$$= 8 \int \left(1 - 2 \sin \theta + \frac{1 - \cos 2\theta}{2} \right) d\theta$$

Attempt to write $\sin^2 \theta$ in terms of $\cos 2\theta$

M1

$$= 8 \left[\theta + 2 \sin \theta + \frac{\theta}{2} - \frac{\sin 2\theta}{4} \right] (+c)$$

Correct integration ft wrong coeffs

A1F

$$8 \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (1 - \sin \theta)^2 d\theta = 8 \times \left[\frac{3\theta}{2} + 2 \cos \theta - \frac{\sin 2\theta}{4} \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$= 8 \times \left\{ \frac{3\pi}{4} - \left(\frac{3\pi}{12} + 2 \cos \frac{\pi}{6} - \frac{1}{4} \sin \frac{2\pi}{6} \right) \right\}$$

$$F\left(\frac{\pi}{2}\right) - F\left(\frac{\pi}{6}\right) \quad \text{OE for relevant area(s)}$$

m1

$$= 8 \times \left(\frac{\pi}{2} - \sqrt{3} + \frac{\sqrt{3}}{8} \right) \quad \{= 4\pi - 7\sqrt{3}\}$$

ft one slip; accept terms in π and $\sqrt{3}$ left unsimplified

A1F

Shaded area = Area of triangle OPQ –

$$2 \times \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} [4(1 - \sin \theta)]^2 d\theta$$

OE

M1

$$\sqrt{3} - 16 \left(\frac{\pi}{2} - \sqrt{3} + \frac{\sqrt{3}}{8} \right) = 15\sqrt{3} - 8\pi$$

Shaded area =

CSO Accept $m = 15, n = -8$

A1

©2025 Exam Papers Practice. All rights reserved.

11

[16]

Q4.

(a) (i) $x^2 + y^2 = r^2, x = r \cos \theta, y = r \sin \theta$

B1 for one stated or used

B2,1,0

$$r^2 = 2r(\cos \theta - \sin \theta)$$

M1

$$x^2 + y^2 = 2(x - y)$$

ACF

A1

4

(ii) $(x - 1)^2 + (y + 1)^2 = 2$

M1A1F

Centre (1, -1); radius $\sqrt{2}$

A1F

3

(b) (i)
$$\text{Area} = \frac{1}{2} \int (4 + \sin \theta)^2 d\theta$$

Use of $\frac{1}{2} \int r^2 d\theta$.

M1

$$= \frac{1}{2} \int_0^{2\pi} (16 + 8 \sin \theta + \sin^2 \theta) d\theta$$

Correct expn of $[4 + \sin \theta]^2$

B1

Correct limits

B1

$$= \int_0^{2\pi} (8 + 4 \sin \theta + 0.25(1 - \cos 2\theta)) d\theta$$

Attempt to write $\sin^2 \theta$ in terms of $\cos 2\theta$

M1

$$= \left[8\theta - 4 \cos \theta + \frac{1}{4} \theta - \frac{1}{8} \sin 2\theta \right]_0^{2\pi}$$

Correct integration ft wrong coefficients

A1F

$$= 16.5\pi$$

CSO

A1

6

- (ii) For the curves to intersect, the eqn
 $2(\cos \theta - \sin \theta) = 4 + \sin \theta$
must have a solution.

Equating rs and simplifying to a suitable form

M1

$$2 \cos \theta - 3 \sin \theta = 4$$

$$R \cos(\theta + \alpha) = 4.$$

OE. Forming a relevant eqn from which valid explanation can be stated directly

M1

where $R = \sqrt{2^2 + 3^2}$ and $\cos \alpha = \frac{2}{R}$

OE. Correct relevant equation

A1

$\cos(\theta + \alpha) = \frac{4}{\sqrt{13}} > 1$. Since must have $-1 \leq \cos X \leq 1$ there are no solutions of the equation $2(\cos \theta - \sin \theta) = 4 + \sin \theta$ so the two curves do not intersect.

Accept other valid explanations.

E1

4

(iii) Required area = answer (b)(i) $- \pi(\text{radius of } C_1)^2$

M1

$$= 16.5\pi - 2\pi = 14.5\pi$$

Ft on (a)(ii) and (b)(i)

A1F

2

EXAM PAPERS PRACTICE [19]

Q5.

©2025 Exam Papers Practice. All rights reserved.

(a)
$$\text{Area} = \frac{1}{2} \int \left(1 + 6e^{-\frac{\theta}{\pi}} \right) d\theta$$

Use of $\frac{1}{2} \int r^2 d\theta$

M1

$$= \frac{1}{2} \int_0^{2\pi} \left(1 + 12e^{-\frac{\theta}{\pi}} + 36e^{-\frac{2\theta}{\pi}} \right) d\theta$$

Correct expansion of $\left(1 + 6e^{-\frac{\theta}{\pi}} \right)^2$

B1

Correct limits

B1

$$= \frac{1}{2} \left[\theta - 12\pi e^{-\frac{\theta}{\pi}} - 18\pi e^{-\frac{2\theta}{\pi}} \right]_0^{2\pi}$$

Correct integration of at least two of the

three terms 1 , $p e^{-\frac{\theta}{\pi}}$, $q e^{-\frac{2\theta}{\pi}}$

m1

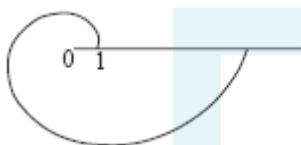
$$= \pi(16 - 6e^{-2} - 9e^{-4})$$

ACF

A1

5

(b)



Going the correct way round the pole

Increasing in distance from the pole

B1

B1

End-points $(1, 0)$ and $(e^2, 2\pi)$

Correct end-points

B1 for each pair or for 1 and e^2 shown on graph in correct positions

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

B2,1,0

4

(c) $e^{\frac{\theta}{\pi}} = 1 + 6e^{-\frac{\theta}{\pi}}$

Elimination of r or θ $\left[r = 1 + \frac{6}{r} \right]$

M1

$$\left(e^{\frac{\theta}{\pi}} \right)^2 - e^{\frac{\theta}{\pi}} - 6 = 0$$

Forming quadratic in $e^{\frac{\theta}{\pi}}$ or in $e^{-\frac{\theta}{\pi}}$
or in r . $[r^2 - r - 6 = 0]$

m1

$$\left(e^{\frac{\theta}{\pi}} - 3 \right) \left(e^{\frac{\theta}{\pi}} + 2 \right) = 0$$

OE

m1

$$e^{\frac{\theta}{\pi}} > 0 \text{ so } e^{\frac{\theta}{\pi}} = 3$$

Rejection of negative 'solution' PI
[r = 3]

E1

Polar coordinates of P are (3, $\pi \ln 3$)

A1

5

[14]

Q6.

(a) $r^2 \sin \theta \cos \theta = 8$

$\sin 2\theta = 2 \sin \theta \cos \theta$ used

M1

$$x = r \cos \theta \quad y = r \sin \theta$$

Either one stated or used

M1

$$xy = 4, \quad y = \frac{4}{x}$$

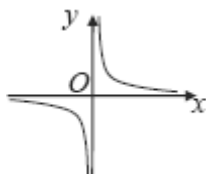
©2025 Exam Papers Practice. All rights reserved.

Either OE eg $y = \frac{8}{2x}$

A1

3

(b)



B1

1

(c) $r = 2 \sec \theta$ is $x = 2$

B1

Sub $x = 2$ in $xy = 4 \Rightarrow 2y = 4$

M1

In cartesian, $A(2, 2)$

$$\Rightarrow \tan \theta = \frac{y}{x} = 1 \Rightarrow \theta = \frac{\pi}{4}$$

Used either $\tan \theta = \frac{y}{x}$ or $r = \sqrt{x^2 + y^2}$

M1

$$\Rightarrow r = \sqrt{x^2 + y^2} = \sqrt{8}$$

$$\theta = \frac{\pi}{4}; r = \sqrt{8}$$

r must be given in surd form

Altn2: Eliminating r to reach eqn. in $\cos \theta$

and $\sin \theta$ only (M1) $\theta = \frac{\pi}{4}$ (A1)

Substitution $r = 2 \sec \left(\frac{\pi}{4} \right)$ (m1)

$r = \sqrt{8}$ (A1) OE surd

Altn3: $r \sin \theta = 2$ (B1)

Solving $r \cos \theta = 2$ and $r \sin \theta = 2$ simultaneously (M1)

$\tan \theta = 1$ or $r^2 = 2^2 + 2^2$ (M1)

$\theta = \frac{\pi}{4}; r = \sqrt{8}$ (A1) need both

©2025 Exam Papers Practice. All rights reserved.

A1

4

[8]

Q7.

(a) $\theta = 0, r = 5 + 2 \cos 0 = 7$ {A lies on C}

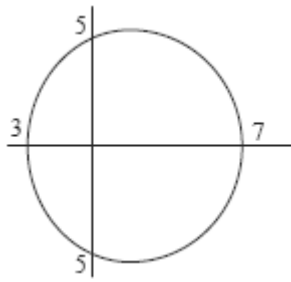
B1

$\theta = \pi, r = 5 + 2 \cos \pi = 3$ {B lies on C}

B1

2

(b)



Closed single loop curve, with (indication of) symmetry

B1

Critical values, 3, 5, 7 indicated

B1

2

(c)
$$\text{Area} = \frac{1}{2} \int (5 + 2 \cos \theta)^2 d\theta$$

Use of $\frac{1}{2} \int r^2 d\theta$

M1

$$= \frac{1}{2} \int_{-\pi}^{\pi} (25 + 20 \cos \theta + 4 \cos^2 \theta) d\theta$$

OE for correct expansion of $(5 + 2 \cos \theta)^2$

B1

For correct limits

B1

$$= \frac{1}{2} \int_{-\pi}^{\pi} (25 + 20 \cos \theta + 2(\cos 2\theta + 1)) d\theta$$

Attempt to write $\cos^2 \theta$ in terms of $\cos 2\theta$

M1

$$= \frac{1}{2} [27\theta + 20 \sin \theta + \sin 2\theta]_{-\pi}^{\pi}$$

Correct integration ft wrong non-zero coefficients in $a + b \cos \theta + c \cos 2\theta$

A1F

$$= 27\pi$$

CSO

A1

6

- (d) Triangle OBQ with
 $OB = 3$ and angle $BOQ = \alpha$
PI

B1

$$OQ = 5 + 2 \cos(-\pi + \alpha)$$

OE

M1

$$\text{Area of triangle } OQB = \frac{1}{2} OB \times OQ \sin \alpha$$

Dep. on correct method to find OQ

m1

$$= \frac{3}{2} (5 - 2 \cos \alpha) \sin \alpha$$

CSO

A1

4

[14]

Examiner reports

Q1.

Many students struggled with this graph sketch, failing to notice that a rearrangement of the equation produces $r \cos \theta = a$ and hence $x = a$, which is a vertical line through $(a, 0)$. Many students plotted some points — which rarely produced the correct answer — whilst others sketched a seemingly random curve, a cardioid being the most popular choice.

Q3.

Part (a) was generally answered well with most candidates forming an equation in either $\sin \theta$ or r and solving it correctly to give the coordinates of P and Q . Candidates who tried to verify the coordinates of P generally failed to even verify them in both polar equations. Part (b) was the most demanding question on the paper. The majority of candidates

scored at least 4 of the 11 marks by applying the formula $A = \frac{1}{2} \int r^2 d\theta$ to find an area of a region partly bounded by the curve C and in doing so they correctly expanded $(1 - \sin \theta)^2$, wrote $\sin^2 \theta$ in terms of $\cos 2\theta$ and integrated correctly. The errors occurred because many candidates did not use the correct limits. A significant number of candidates also failed to find, or even consider the need for, the area of triangle OPQ and so correct answers for the area of the shaded region R were generally only presented by the most able candidates.

Q4.

This question on polar coordinates proved to be the most demanding question on the paper. In part (a), the better candidates had no problems in finding the cartesian equation for C_1 and rearranging it by completing the squares so as to be able to deduce that the curve was a circle. Many other candidates failed to eliminate θ even after a page or more of working and abandoned part (a) of the question.

Many candidates scored heavily on the more familiar part (b)(i), finding the area bounded by the second curve C_2 .

Proving that the two curves did not intersect in part (b)(ii) was a challenge, even for the better candidates, and it was rare to award all four marks for this part of the question. However, some excellent solutions were seen which used a variety of methods including forming and solving quadratic equations in either $\sin x$, $\cos x$, $\tan x$, $\sec x$, use of calculus and use of the R, α form.

In the final part of the question, those who used part (a)(ii) to find the area enclosed by C_1 and used it with their answer to part (b)(i) had no problems scoring both marks. Those who tried to use integration to find the area enclosed by C_1 did not analyse the problem fully and failed to score. A common misinterpretation of the word 'intersect' is indicated by the not uncommon answer 'Since C_1 and C_2 do not intersect the required area is just 16.5π , the area bounded by C_2 .'

Q5.

Some candidates integrated r instead of r^2 to find the area of the shaded region. If they

had first stated the formula $A = \frac{1}{2} \int r^2 d\theta$, which is given in the formulae booklet, before substituting for r further credit could have been awarded. The two most common errors were incorrectly squaring $6e^{-\frac{\theta}{\pi}}$ to get $6e^{-\frac{2\theta}{\pi}}$ and integrating $12e^{-\frac{\theta}{\pi}}$ incorrectly to get $-\frac{12}{\pi} e^{-\frac{\theta}{\pi}}$. The quality of sketches varied significantly with some even starting at the pole. A significant minority of candidates either just gave the coordinates of one end point, missing $(1, 0)$ or gave a decimal approximation for e^2 or gave the incorrect answer $(e^2, 0)$. It was surprising to find some candidates giving the coordinates in reverse order. In part (c), most candidates formed a correct equation by equating the r terms but in general only the better candidates were then able to form and solve the resulting quadratic equation in $e^{\frac{\theta}{\pi}}$. Only a small number of candidates failed to reject the negative value for $e^{\frac{\theta}{\pi}}$ before going on to get the correct coordinates for the point P . Again candidates should use exact forms and not give a decimal approximation in place of $\pi \ln 3$. A common mistake seen in solving ' $e^{\frac{\theta}{\pi}} = 1 + 6e^{-\frac{\theta}{\pi}}$ ', is illustrated by ' $\frac{\theta}{\pi} = \ln 1 + 6\left(-\frac{\theta}{\pi}\right)$ '.

Q6.

Those candidates who replaced $\sin 2\theta$ by $2\sin \theta \cos \theta$ generally obtained the correct cartesian equation in part (a). The sketch of the curve C (rectangular hyperbola) required in part (b) was not answered as well as expected with many sketches consisting of closed loops. Candidates presented a variety of acceptable methods for part (c). Those who eliminated r were required to obtain a trigonometrical equation in a single angle before any mark was awarded. Usually candidates who had found the correct equation went on to obtain the correct exact values for the polar coordinates of A . The second most popular method involved working with the cartesian form of the equation of C . Candidates who recognised the cartesian form of the equation of the line equivalent to the polar form given in part (c) of the question generally had no difficulty getting the cartesian coordinates for A as $(2, 2)$, but then a significant minority could make no further progress.

Q7.

Parts (a) and (b) were generally answered well, but some candidates did not identify the critical value of 5 on their sketches. In part (c), although the usual errors were seen — for example the wrong expansion $(5 + 2 \cos \theta)^2 = 25 + 10\cos\theta + 4\cos^2\theta$ or a sign error in the identity for $4\cos^2\theta$ in terms of $\cos 2\theta$ — most candidates had a thorough understanding of the method required to find the area of the region bounded by the curve. Although only a minority of candidates scored full marks for the final part of this last question, it is pleasing to report that a significant number of other candidates were awarded partial credit for finding an expression for OQ , although many lost at least one mark because they used $\pi - \alpha$ instead of $\alpha - \pi$ for θ . The most common wrong method seen involved the integral

$$\int_{-\pi+\alpha}^{\pi} \frac{1}{2} r^2 d\theta, \text{ for which no credit was awarded.}$$